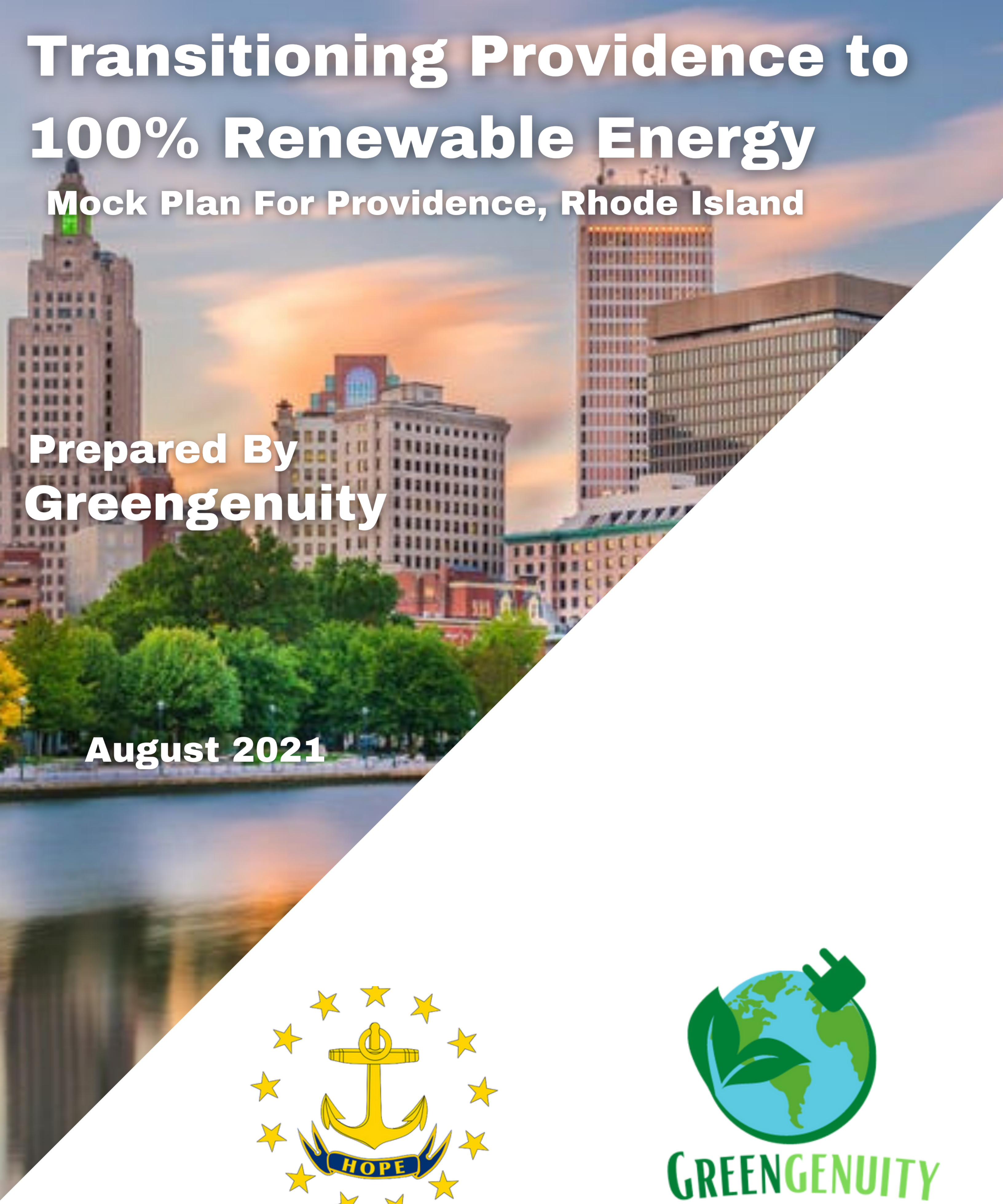
Transitioning Providence to 100% Renewable Energy

Mock Plan For Providence, Rhode Island



Prepared By

Greengenuity

August 2021

GREENGENUITY

Table of Contents

I. Introduction 3

A. Who Are We?.. 3

B. What Do We Do?.. 4

II. Layout of Our Service 4

III. Overview of the Plan.. 5

A. Providence Energy Consumption.. 5

B. Solar Energy... 5

C. Wind Energy... 5

D. Biofuel.. 6

E. Energy Storage.. 6

F. Transmission... 6

G. Demand-Side Reduction... 6

IV. About Providence 6

A. Electricity Usage and Sources.. 6

B. Rates.. 7

C. Current Goals.. 7

D. Existing Renewables and their Megawatt Capacity...................... 8

V. Rhode Island Renewable Energy Programs & Funds........................ 8

VI. Solar.. 9

A. What is Solar Energy?.. 9

B. How much energy will we be getting from solar? What is the cost, size, and locations?....... 9

C. Solar Energy Programs & Incentives.. 10

D. Providence’s Architectural Data for Solar Panel Data................ 10

VII. Wind 11

A. Revolution Wind Project........ 11
B. Offshore Wind in Providence Residencies........ 12
C. Large Offshore Wind Projects........ 12
D. Onshore Wind........ 12
E. Cost........ 13

VIII. Biofuels........ 13
A. What is Biofuel?........ 13
B. Making Biofuel........ 13
C. How it Works........ 13
D. Recommendations........ 14
E. Costs & Implementation........ 14

IX. Energy Storage........ 14
A. Liquid Metal Batteries........ 14
B. Costs and Implementation........ 15
C. Other Batteries........ 15

X. Demand-Reduction Methods for Providence........ 16
A. Installing Roof Solar Pannels........ 17
B. Improving Homes/Buildings Insulation........ 17
C. Avoiding Over-Lighting and Using Natural-Lighting........ 17
D. Smart Building Management........ 17

XI. 2-D Model........ 18
A. What is in the Model........ 18
B. Why the Model is Used........ 19

XII. Partnerships........ 19
A. American Superconductor (WindTec - AMSC)........ 19
B. SolSmart........ 20
C. First Solar........ 20
D. Ambri........ 20

E. Verdigris..20

XIII. References... 21

I. Introduction

A. Who Are We?

Greengenuity is an organization composed of hardworking, passionate students, supported by an extensive network of coaches, industry partners, and government leaders. We were founded as part of PrepareRI's 2021 Innovation Challenge. All of us are dedicated to solving the drawbacks of renewable energy and helping guide our world to a more sustainable future.

Here is our team:

- **Leadership**:
  - Paula Zhuang, CEO
  - Joshua Rodriguez, COO
  - Lyneth Mendoza, CFO
  - Monta Crooks, CIO
  - Tireni Onasanya, CHRO

- **Business and Finance**:
  - Alyn Rodriguez
  - Nathan Dieterich
  - Margaret MacKechnie

- **Marketing and Communications:**
  - Ava Marchand
  - Jayden Castro
  - Mae D'Ambra
  - Tyler Robichaud

- **Research and Development:**

- Alan De La Cruz
- Braden DeGraw
- Darren Ok
- Ginger Taylor
- Shaili Sullivan

B. What Do We Do?

Greengenuity is mainly a consulting firm, partnering with municipal governments to provide them with comprehensive, specific plans to transition to 100% renewable energy. Our plans focus on installing grid-level storage, such as liquid metal batteries and hydroelectric pumped-storage, as well as building renewable energy sources to feed into the city's electric grid. All recommended infrastructure will be within or close to the city to improve the local economy and improve reliability. Sources may include solar, wind, geothermal, biofuel, hydropower, hydrogen, and tidal energy. In order to develop the plan, we will research the infrastructure, geography, and current renewable energy resources of the city, deciding on the optimal locations to install batteries and energy sources while making only minimal changes to existing infrastructure.

To provide physical components and installation (such as batteries) at a lower rate, we partner with renewable energy corporations—mostly local—and provide their services to municipalities. However, cities are not required to use these services.

To model our plan, we created an interactive 2-D model on ArcGIS. Our website allows each city to plan out the layout of proposed installations, as well as simulate the underlying electrical grid. Furthermore, the simulation can predict the optimal power sources for an area. Data may include how much electricity comes from each source, what uses the most energy, how much energy is stored in the batteries, and more. In addition, it will draw our attention to any maintenance that has to be done along the way.

The customer cities pay an upfront fee for our completed plan. In addition, when cities pay our partnered companies for construction, materials, and setup, we follow a revenue-share model by taking a portion of those profits. Finally, we charge cities a small recurring fee for updates to the ArcGIS model, continual updates on research and how that would affect the city's 100% plan, and general supervision of the installations.

II. Layout of Our Service

We plan to compile solar, wind, and biofuel energy with our specific energy storage systems to power Providence with 100% renewable energy. The following is our methodology and prescribed plan:

1. Research and evaluate the city's current geography, infrastructure, climate, policies, and energy consumption (completed within this packet).
2. Establish the current renewable energy structures participating in Providence's energy consumption and what sources and structures we would like to add. Our plan for Providence will utilize a combination of three different renewable energy sources: solar, wind, and biofuels (completed within this packet).
3. Create an ArcGIS model to test show our modifications to the Providence energy grid will affect its current state. The model allows us to map out where we want to locate the different infrastructure and how it will literally appear in the city (linked in this packet).
4. Present to Providence government officials and discuss variables such as cost and timeline.
5. Utilize local contractors (solar panel installation companies, wind turbine manufacturers, etc.) in order to begin the installation process.

III. Overview of Our Plan

A. Providence Energy Consumption

As an estimate, Providence uses around 1,305,102 MWh annually. Taking into account the solar farm in Johnston that provides Providence with renewable electricity through virtual net metering (and is local enough to provide relative security and help with the economy) along with the existing renewables in Providence, 1,237,495 MWh of annual electricity still needs renewable generation. Our plan is to generate 40% of that with solar (494,998 MWh annually), 40% of that with wind (494,998 MWh annually), and 20% of that with biofuels (247,499 MWh annually).

B. Solar Energy

First, solar panels are an ideal renewable source for this city due to the ease of installation and the economic benefits of job creation. These would include small-scale installations on residential buildings and private corporations, which a greater incentivized program for solar panels would accomplish, as well as large-scale municipally-controlled systems on areas like brownfields. Unlike wind turbines, solar panels are compact, quiet, and generally unobtrusive and are therefore ideal for placement in urban environments, where space is at a premium.

C. Wind Energy

Additionally, wind generation is a clear choice. Offshore wind power is not only one of the most commercially developed renewables in RI, but it is a great option for Providence based on its geographical location near the water. Although on-shore wind power takes up more valuable land and skyline, it can be slightly less expensive, so we are including that as well.

D. Biofuel

Finally, biofuel plants can be developed within Providence, and some are already in the process. Unlike solar or wind, biofuel plants can be built almost anywhere there is sufficient space, as biofuel can be transported from other areas and does not need to be generated on site. Proposed biofuels include algae and sewage.

E. Energy Storage

As for 15-hour energy storage, about 894 MWh capacity would be needed. As a price estimate, Tesla Megapack lithium-ion batteries would cost $255,993,940. However, liquid metal batteries are an alternative that would be more cost-effective in the future as manufacturing processes evolve. Liquid metal batteries also last for significantly longer and thus require less maintenance than traditional lithium-ion batteries. However, their high operating temperature means that they will have to be installed in out-of-the-way places, most likely underground.

F. Transmission

To transport electricity from renewable energy sources to batteries to consumers and convert the energy between AC and DC, existing hardware (e.g. power lines) can be retrofitted to accommodate the needs of the new additions. However, new installations may be necessary.

G. Demand-Side Reduction

Reducing the load on the demand side and understanding electricity usage is an essential part of our solution, as this introduces more reliability into the grid. This is especially true if renewable sources are the main sources of electricity; for example, reducing electricity usage during times of low generation, like at night, would assist with stability. Overall, less usage and the ability to predict how much electricity will be used on certain days and at certain hours is invaluable. Thus, we provide a few different recommendations for Providence to consider on the energy conservation side.

IV. Providence

A. Electricity Usage and Sources

The City of Providence consumes about 1,305,102,738 kWh of electricity. This number was estimated from the per capita electricity usage in RI multiplied by the number of people in Providence. Though the exact number varies from year to year. Currently, the vast majority of Providence's power is provided via natural gas, through either Direct Energy or National Grid.

B. Rates

The electricity usage throughout Providence is divided into three categories; commercial, residential, and industrial. The average commercial electricity rates in the city are 8.1¢/kWh. This is 31.76% less than the average rate in RI, which is 11.87¢/kWh. It is also 19.72% less compared to 10.09¢/kWh, which is the national rate. The residential electricity rates average is 14.24¢/kWh. The rate in Providence is 1.11% less than the Rhode Island average rate of 14.4¢/kWh. It is also 19.87% greater than the national average rate of 11.88¢/kWh. Switching to renewable power for electricity would ultimately decrease electricity prices for residents. Last is the industrial electricity in Providence. The average industrial electricity rate in Providence is 5.25¢/kWh. This industrial rate in Providence is 50.84% less than the Rhode Island average rate of 10.68¢/kWh. It is also 21.29% less compared to the national average rate, which is 6.67¢/kWh.

C. Current Goals

Rhode Island aiming for 100% of municipal buildings' electricity to be renewable by 2030. By 2040, this will include heating systems. In 2035, 50% of citywide electricity will be carbon-free. Finally, by 2050, 100% of citywide electricity will be carbon-free. These are ambitious plans, and we intend to help facilitate the transition.

In addition, Providence hopes to gain status as a city that cares about the environment. In fact, Providence was awarded a Silver SolSmart Designation, earning high-profile national recognition for the steps it has taken to reduce barriers to adopting solar energy. Providence sees equity in renewable energy as a crucial goal, and Providence officials emphasize the importance of helping low-income families and communities take advantage of renewable sources to lower their energy costs, since initiatives like residential solar may not be feasible in these neighborhoods. Thus, one of the focus points of Greengenuity's plan is to install renewable infrastructure in communities that need it most, allowing it to directly help them and decrease their living costs.

Moreover, strengthening the Providence economy as a whole is a goal that was highlighted by many city and state employees. This includes the creation of jobs, the bolstering of renewable industries, and the support of local businesses. Greengenuity keeps all of this in mind, selecting local businesses for partnerships when possible while still offering lower prices to Providence. The installations that we recommend will not only increase local employment but also help these businesses.

Finally, it is important to the City of Providence to secure their resiliency to extreme weather and grid blackouts, events that become more likely as climate change accelerates. The in-city battery storage hubs that Greengenuity recommends will ensure that the citizens of Providence always have power, no matter what issues the national grid comes across. This microgrid-like system protects Providence.

D. Existing Renewables and their Megawatt Capacity

Wind power in Providence (all on-shore wind):

- NBC Field's Point Wind Farm (4.5MW)
  - Annual production: 1,468,070.25 kWh
- WED Green Hill (3MW)
  - Annual production: 6,570,000 kWh (estimate, using 25% capacity factor)
- WED Plainfield 1-3 (3MW each)
  - Annual production: 19,710,000 kWh (estimate, using 25% capacity factor)
- WED Shun 1-3 (3MW)
  - Annual production: 6,570,000 kWh (estimate, using 25% capacity factor)

Photovoltaic solar power sources in Providence or providing power to Providence:

- Virtual Net Metering (e.g., from Johnston Solar, 3.9MW)
  - Annual Production: 30,500,000 kWh (estimate)
- Little Bay (1.3MW)
  - Annual Production: 2,790,000 kWh (estimate)

There are no biomass plants (created from landfill gas) in Providence.

V. Rhode Island Renewable Energy Programs & Funds

There are many resources Providence has access to that can further their goal of running on 100% renewable energy. Rhode Island Commerce's Renewable Energy Fund (REF) aids in increasing the use of renewable energy and the diversity of renewables throughout Rhode Island. The REF provides grants for renewable energy projects that have the potential to produce electricity in a cleaner, more sustainable manner, while stimulating job growth in the green technology and energy sectors. The REF funds programs such as Commercial Scale Projects, Small Scale Projects, Community Renewables, and Brownfields Solar PV Program.

The Rhode Island Efficient Buildings Fund (RIEBF) is a revolving fund that provides governmental entities with low-interest rate loans on energy efficiency and renewable energy projects that exceed the annual debt service on the annual energy savings achieved by the projects.

The Rhode Island Infrastructure Bank's Commercial Property Assessed Clean Energy (C-PACE) is a financing program for energy efficiency and renewable energy projects in commercial and industrial properties. Projects surrounding energy efficiency and renewable energy are eligible for 100% C-PACE financing. Buildings that are eligible for this program include office buildings, manufacturing facilities, agricultural lands, nonprofit organizations, and multifamily homes (5+ units).

The Renewable Energy Growth Program (REG Program) is administered program provided by National Grid that supports the development of distributed generation projects in Rhode Island including: technologies include wind, solar, hydropower, and anaerobic digestion.

Brownfields Area-Wide Planning Program assists communities in responding to challenges caused by local brownfields, which are contaminated sites unfit for development. Particularly, where multiple of these sites exist in close proximity, connected by infrastructure, and limit the economic, environmental and social prosperity of their surroundings. This program enhances EPA's core brownfields assistance programs by providing grant funding to communities so they can perform the research needed to develop an area-wide plan and implementation strategies for brownfields assessment, cleanup, and reuse.

VI. Solar

A. What is Solar Energy?

Solar power is energy from the sun that is converted into thermal or electrical energy. Solar energy is the cleanest and most abundant renewable energy source available, and the U.S. has some of the richest solar resources in the world. Solar technologies can harness this energy for a variety of uses, from electricity generation, to providing interior lighting, and water heating for domestic, commercial, and industrial use.

Photovoltaics (PV) and concentrating solar-thermal power (CSP) are the two main forms of Solar energy. Photovoltaics is the most common as it utilizes solar panels to absorb sunlight to generate electricity. Concentrating solar-thermal power utilizes mirrors that reflect and concentrate sunlight onto receivers that harnesses the solar energy to generate heat, which can either be stored, or, used to generate electricity.

B. How much energy will we be getting from solar? What is the cost, size, and locations?

Solar power is more affordable, accessible, and prevalent in the US than ever before. From 0.34 gigawatts in 2008, US solar power capacity has grown to 97.2 gigawatts. This is enough to power more than 18 million homes. Over 3% of United States' electricity comes from solar energy. Rhode Island as a whole would need **about 4,440 gigawatt-hours** of renewable-energy generation to meet its predicted 7,700 gigawatt-hours of electricity use in 2030. That works out to securing about 440 gigawatt-hours of new renewable power annually for 10 years. For Providence, 494,997,867.2 kilowatt-hours more of annual power needs to be installed from solar. Using the estimate from Freedom Solar Power of $200,000 for a solar system that produces 136,000 kilowatt-hours per year, this would be a cost of about $727,938,040. This would require about 2,390,960m^2 of space, or 593.6 acres. In Rhode Island, the cost of solar panels installation averages from $12,580 to $17,020 with the average cost per watt being from $2.52 to $3.40. Providence offers no-cost energy assessment for its members that ensures the best value for going solar. Depending on the city's decision, these could be placed on brownfields, rooftops, community buildings, fields, or other open areas. Residential solar and private commercial solar could also contribute to this number.

C. Solar Energy Programs & Incentives

Community solar projects allows for a community to have access to solar energy if they cannot physically obtain solar technology. The solar energy is generated by a community solar farm, and serves multiple households at a time. By participating in community solar projects, members contribute to clean energy use while saving money on their monthly electricity bill. Through the Renewable Energy Fund, there is an available incentive program called the Community Renewables Program for community solar developers who will pass on the grants to subscribers in the form of a bill credit on their community solar bill. This program will allow Low-to-Moderate-Income (LMI) recipients on the National Grid A-60 rate to receive a flat rebate grant amount of $500 per subscriber and Basic residential recipients on the National Grid A-16 rate to receive a flat rebate grant amount of $300 per subscriber.

Clean Energy Grant Incentive Program is a program aimed to decrease investment costs for carbon-free solar energy installations and increase the use of renewable energy resources in State Agencies, public colleges and universities.

D. Providence's Architectural Data for Solar Panels Data

Total Potential refers to the entire quantity of solar possible,

Technical Potential is a subset of total potential that includes only areas that are suitable for solar

Economic Potential is an estimate of the solar potential that is likely to be installed, given the current cost of the technology, the current financial incentives available, and the household economics specific to a municipality.

Rooftop Solar

Number of Rooftops	Total potential (MW)	Technical potential (MW)	Low economic potential (MW)	High economic potential (MW)

36,200	355.7	103.1	5.6	12.2

Brownfields

Number of Brownfields	Total potential	Low Technical potential (MW)	High technical potential (MW)
164	79.8	0.0	22.9

Residential buildings that cannot host solar PV

Total Number of residential rooftops (Thousands)	Fraction of Area with no buildable area	Number of rooftops with no buildable area	Economic Potential (MW)
30.7	3%	926	5.6

VII. Wind

Rhode Island wind turbines produce approximately 40,000 megawatt-hours (MWh) per year out of the 7,700,000 MWh of energy consumption the state makes annually. We aim to produce 40% of Providence's energy consumption, 3,850,000 MWh annually, to wind energy. Installing wind power consists of the purchasing of turbines, land, and parts, as well as the upkeep of the system. Ultimately, operation and maintenance of wind turbines requires insurance, land rent, service, repair, spare parts, administrative tasks, power needed to maintain the turbines, and other miscellaneous facets.

A. Revolution Wind Project

Over 95% of wind energy in Rhode Island is located offshore. In 2016, the nation's first 30 MW offshore wind project was built off the coast of Block Island, Rhode Island. Companies Ørsted and Eversource are partnering to expand offshore wind into the Port of Providence. The project, Revolution Wind, is a 704MW proposal that will supply Rhode Island with 400MW of energy and Connecticut with 304 MW of energy annually. The wind farm will cover 256 square miles of federal waters between Block Island, Rhode Island, and Martha's Vineyard. The companies also made a PLA with Rhode Island Building and Construction Trades Council and Dimeo Construction to ensure that all service within the plant will be completed by local union labor; therefore, more jobs will be brought to the area. The Revolution Wind project could begin construction as early as 2023.

B. Offshore Wind in Providence Residencies

A 6MW offshore wind turbine in Rhode Island can power over 4,000 homes annually, assuming a 48% capacity factor and an average household use of 500MWh per month or 6,000 KWh per year. Rough calculations will also allow us to assume that one 6MW turbine will produce 52,560 MWh annually. Taking our calculations into account, because there are approximately 63,000 occupied homes in Providence as of 2020, the city would need 90 6MW offshore turbines to fully power all residencies. To power all of providence, about 118 of these turbines would be needed. This could potentially be accomplished, since offshore wind farms are nothing new to Rhode Island. In fact, the addition of 17 more offshore wind farms equivalent to the Block Island Wind Farm could roughly generate the amount of energy needed to power all 63,000 of Providence residencies. However, we must account for offshore wind's unreliability, since the turbines cannot always operate at full capacity due to inconsistent wind speeds. To counteract this inherent instability, we will use batteries to store energy while the system is producing power, using the batteries to fill in any gaps in the power supply.

C. Large Offshore Wind Projects

Companies such as General Electric and Vestas Wind Systems have recently built and proposed the largest and most-efficient offshore wind turbines in the world. Vestas' new V236 turbine, which is planned to be built in 2022, will be 855 feet tall, with 380-foot-long blades, and the largest output yet of a singular turbine, 15MW. General Electric's Haliade-X turbine, which has already been built, has an output of 14MW, 813 feet tall, with 351-foot-long blades. Vestas's turbine plans to produce 80GWh/year, and General Electric's turbine will produce 74GWh/year. According to this, this wind turbine could produce 74 out of the 495GWh per year needed to cover 40% of Providence's annual energy consumption. Although these turbines can produce an overwhelming amount of energy, due to the cost of building such turbines, and the rarity of these new turbines, the price isn't feasible for Providence. However, the mass push of development for wind turbines is creating more and more efficient options.

D. Onshore Wind

A typical 1.5 MW onshore wind turbine in Rhode Island can power approximately 440 homes annually, assuming a 20% capacity factor and an average household use of 500 kWh. Additionally, an average turbine with a 2.5-3 MW capacity can produce up to 6 million kWh per year. Turbines of this size are larger than most out of the onshore turbine spectrum. However, the installation of 82 2.5-3MW turbines could produce enough annual energy to power 40% of Providence. The only problem Providence faces with onshore turbines is the lack of space. The city itself is not very large to support a surplus of onshore turbines. Due to this spatial disadvantage, Green Development is currently the only company actively pursuing onshore wind turbines in Rhode Island. In 2021, they have planned to develop three more turbines on land owned by Johnson and Wales University and the Port of Providence. These turbines each have a capacity of 1.5MW and produce an average of 43.5 MW of annual onshore wind power.

E. Cost

The 2019 levelized cost of energy (LOCE=2020$/MWh) of offshore wind is $85/MWh and the 2019 levelized cost of energy of land-based wind is $82/MWh. It costs $2.6 – $4 million per average-sized commercial wind turbine. The average cost per megawatt (MW) of electricity-generating capacity is $1.3 million. Most commercial wind turbines have a capacity of 2 to 3 megawatts (MW), however offshore turbines can have a capacity of up to 12 megawatts. The cost of employing fewer, larger turbines rises as the size of the turbine increases. However, there are advantages to using fewer, larger turbines; the complexity and construction of the whole farm site is substantially decreased by using fewer, larger turbines. Therefore, the installation of larger turbines offshore wind farms, particularly in in Port of Providence, would be beneficial.

VIII. Biofuels

A. What Is Biofuel?

Biofuels are combustive fuels made from recently harvested plants. The majority of biofuels used today are derived from agricultural products. Biofuels derived from algae and cellulosic biomass (cornstalks & wheat straw) could provide a sustainable fuel source that does not compete with food and other supplies. These advanced biofuels have the potential to be mass-produced and obtained from CO_2 consuming sources. Using biofuels is one way of reducing dependency on fossil fuels. We also plan to experiment with sewage as a potential biofuel source. However, studies around sewage energy are still in the works. Biomass energy implies generating electricity using biomass as fuel.

B. Making Biofuel

Plant matter is harvested and converted into liquid biofuel. The plant matter is broken down into a "mash" in the case of corn. Enzymes break down the maize “mash” into sugars, which yeast ferments into alcohol and a plant byproduct. The “mash” is filtered to remove the alcohol, and the plant waste is used as livestock feed. The alcohol is converted to ethanol, which can be used as a biofuel, with the last addition of heat.

C. How It Works

Unlike gasoline mixed with ethanol, biofuel demands the usage of a specific vehicle. Vehicles built to run on both gasoline and ethanol mixes are known as flexible-fuel vehicles. Biofuel is typically offered in the United States as a blend of 85% ethanol and 15% gasoline, which permits the ethanol to burn more easily in cold weather.

D. Recommendations

Biofuel is a safer alternative to traditional fossil fuels since the plants used to make it can be cultivated and harvested, making them renewable resources. It still takes a lot of energy to make, so it's not perfect; but it can create 12 to 20 gallons of ethanol for every gallon of oil needed in processing and distributing biofuels. Another option is to use biomass energy in fuel cells, which are capable of efficiently converting hydrogen-rich fuels into power and water while emitting less pollution.

E. Cost & Implementation

Regular diesel engines can be converted to run on biodiesel with very little effort. In the weeks following the transition, rubber gasoline lines must be replaced with metal one and filters must be examined. In other words, a complete engine conversion is required for non-diesel engines. In terms of electricity, the cost of biomass energy per kWh (kilowatt-hour) is roughly 5-10 cents, which equates to USD 1500 – USD 1800 per kWp (kilowatt-peak). Although the cost of electricity generated from biomass energy varies, as it is influenced by the following factors:

- What type of biofuel was used?
- The method used to produce biomass energy from biofuel
- The plant's size and system design for generating biomass energy

The cost of generating electricity from anaerobic digestion of animal manure is between 3.7 and 5.4 cents per kilowatt-hour. The costs of this investment in biomass production range from USD 230,000 to USD 260,000. The heat created by digester gases is utilized to generate energy, while the liquid and fiber digestion wastes are used to fertilize land in an effective way. 247,498,933.6 kWh will need to be generated from biofuel yearly, which necessitates the construction of biofuel plants.

IX. Energy Storage

A. Liquid Metal Batteries

Liquid metal batteries have not been in the market for a long time, but there is a lot of research backing their ability to be the next big thing in renewable energy storage. Currently, the most popular liquid metal batteries manufacturer is Ambri. Liquid metal batteries can provide a fast response for frequency regulation applications while also offering long-duration performance for reserve capacity markets.

One pro to buying liquid metal batteries are that they are cheaper than lithium-ion. Ambri's batteries use electrode materials that cost 33% of those in NMC lithium-ion. Manufacturing costs are also 33%-50% less than lithium as it is much simpler, lowering the capital expense per MWh compared to lithium-ion batteries. These batteries will create less waste product, leave a smaller carbon footprint as it has a longer lifespan, and reduce energy loss through overheating since it can overcharge for extended periods. It costs less up front and lasts longer, and has a lower operating cost over its life relative to lithium-ion.

B. Costs and Implementations

To store 894 MWh, which is about 6 hours of energy for the city of Providence, it would take $75,990,000-$106,386,000 using Ambri's batteries from 2022-2030.

Calculations: The price for 894 MWh of lithium-ion batteries as of right now is $255,993,940. Ambri states that their costs will be 30-50% below the current market prices by 2022-2030, which corresponds to $75,990,000-$106,386,000.

C. Other Batteries

Lithium-Ion Batteries

The most known and popularly used way of battery energy storage is lithium-ion Batteries. The current lithium-ion pricing stands at about $137 per kilowatt-hour and will drop as low as $100 per kWh by 2023. The lowest pricing for lithium-ion batteries is generally available for either a major supply contract or for very large-scale deployments of 500 megawatt-hours and above. Since most projects today are not that large, that $137,000 per MWh figure will be closer to $150,000 to $170,000 per MWh, and perhaps as high as $200,000 to $210,000 per MWh on the battery-pack level, depending on the size of the project. The total project cost is close to $470,000 per MWh which includes energy capacity, power conversion systems, construction, and commissioning.

Although lithium-ion batteries have a large energy capacity, some problems are being faced consisting of protection and aging. Over extended periods of charging and discharging, the batteries will degrade and stop holding charge as effectively (Remis). Once lithium-ion batteries have gone through between 500 to 1,000 cycles, it can no longer be in use. Also, lithium-ion batteries can fail catastrophically, and improper handling or accidents could result in enormous fires (Pros and Cons). Another issue with lithium-ion batteries is disposal. Lithium-ion batteries can be recycled, but this process is energy-intensive and there are not currently facilities capable of handling the volume of batteries that would be required for this project.

Hydroelectric Energy Storage

Used for long term storage, excess energy pumps water into a reservoir and then it is released down a hill through turbines to generate electricity when needed. This type of energy storage requires geographical height and water availability. It has the largest energy capacity with about 95% of all actively tracked storage worldwide being stored by hydroelectric. It has a low maintenance cost and has the lowest levelized cost of electricity (LCOE) out of all the electricity energy storage technologies, at only roughly $165,000 per MWh of storage.

The main drawbacks of this technology are the geographical implications, high upfront cost, and environmental impacts on the local ecosystem. For a pumped hydro storage to be considered, there must be a large difference in elevation between the two storage units. Even after finding an area to construct the storage units, the construction itself usually takes years to complete. The conventional hydro storage unit dam rivers to create a reservoir, which can have an environmental impact as it affects the aquatic ecosystem, while also impacting the local wildlife area.

Compressed Air Energy Storage (CAES)

CAES uses excess energy to compress air using a pump and storing it in a high-pressure container. When energy is needed, the high-pressure air passes through a turbine, generating electricity. CAES plants have a very large power rating and storage capacity, low self-discharge, and a long lifetime. All this is achieved while being one of the most cost-effective storage units at about $105,000 per MWh.

There are many companies also working on the loss of energy through heat by storing that heat and reusing it during decompression. This new system was researched and developed by the European Commission named Advances Adiabatic Compressed Air Energy Storage (AA-CAES). Some problems still arise concerning the sites, as many use old abandoned salt mines or caverns to pump the air into. Although not all areas are suitable for CAES, it’s estimated that 70% of the earth’s landmass can be used to support CAES systems. There also some companies developing a CAES system that uses the ocean’s natural pressure to facilitate storage.

X. Demand-Reduction Methods for Providence

The term "energy conservation" refers to the reduction of energy usage. After oil, gas, coal, and nuclear energy production, it is sometimes referred to as the “fifth fuel”. By reducing energy consumption, less energy will need to be produced, decreasing the load on the system. Energy conservation in building requires using energy-saving building materials in the design and construction of buildings, improving building insulation and sealing performance, and enhancing the efficiency of climate control systems to reduce energy consumption.

These are some of the **most effective**, and **well-proven** energy saving measures to consider in order to **lower energy consumption:**

A. Installing More Efficient Systems

Older devices generally require significantly more power than modern ones. Modern devices are usually built with efficiency in mind, and are designed to keep power use to a minimum. In addition to more efficient construction, many modern devices include smart systems to use less power. For instance, only running air conditioning while there is someone inside the house. Additionally, all devices degrade over time, and older devices will inherently lose performance and efficiency over time. Replacing older devices could significantly reduce electricity use.

B. Improving Homes/Buildings Insulation

Heating and cooling often accounts for a large amount of home electricity usage. One way of decreasing this cost, while maintaining a comfortable home temperature, is to install more efficient insulation. Additionally, if an energy-efficient space heater is chosen, one will be able to save even more money on heating.

C. Avoiding Over-Lighting and Using Natural-Lighting

Getting rid of, or avoiding to put on lights that aren’t necessarily needed in rooms can provide major energy savings. Over-lighting will not only increase energy bills, but it will also create glare, which may induce eye strain and headaches. Combining the use of natural light with the elimination of excessive artificial illumination will alleviate these issues. The same principle applies to outdoor space. Using a solar lamp post will help light outdoor space without impacting your carbon footprint.

D. Smart Building Management

In order to have reliability in the electric grid with renewable energy, it is important to understand where electricity is being used and figure out how to reduce that during peak demand times. Smart building management is the solution, monitoring electricity usage and seeing where efficiency can be improved. One company highly experienced in offering this is Verdigris, who offers energy meters with AI. The sensors provide information that can then be used for adaptive automation, allowing lower electricity usage while still maintaining the same level of comfort. In addition, the technology’s predictive abilities improve reliability. Thus, it is recommended to install this AI within large, energy-consuming buildings in Providence, cutting down on their usage.

XI. 2D Model

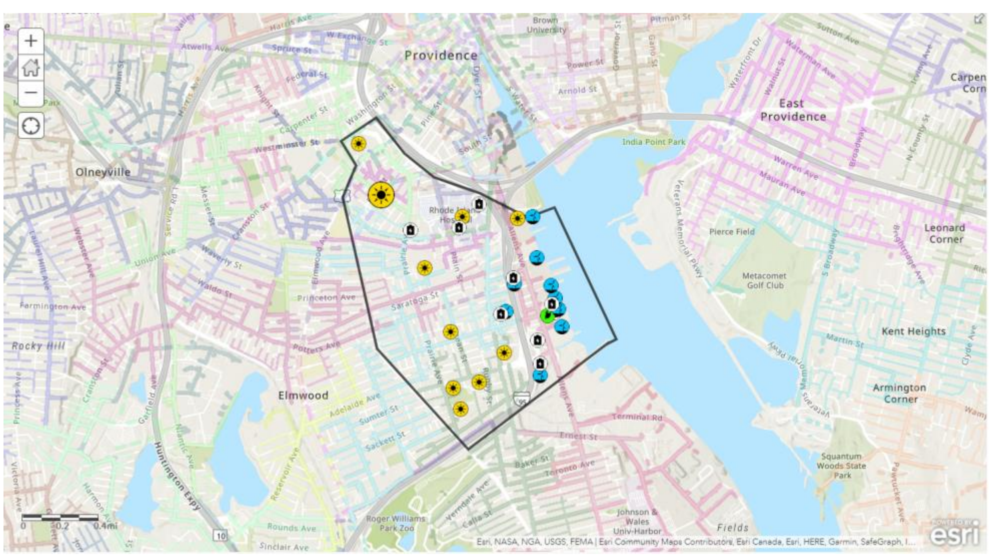
To visualize our plan and the locations of each battery and energy source, Greengenuity has utilized ArcGIS, a geographic information system software created by Esri. The link is here:

A. What is in the Model

As a base for the 2D model, the city of Providence is highlighted as the area we are focusing on. In order to keep the scope of the initial model feasible, the districts of upper and lower South Providence (outlined) were selected as a case study, due to its containing both commercial and residential buildings, port access, and key buildings such as its hospital. Within this layer, National Grid's power distribution lines are depicted, to highlight the coverage of the already existing power grid and to show the power lines that can be used to carry renewable energy to buildings. Next, the locations of pre-existing renewable energies are shown, as well as their power outputs. Finally, the positions of Greengenuity's proposed infrastructure are shown, along with their predicted power output.

There are different icons for solar arrays, wind turbines, biofuel plants, and battery banks. In addition, the model is fully interactive. Each icon is clickable, showing the proposed energy output or energy capacity. This information was calculated using energy data from the city of Providence. These layers can also be turned on or off, allowing a before-and-after view.

Depicted below is an image of what the model looks like:



B. Why the Model Is Used

The 2D model is a visual representation of the proposed solution, which can be modified at any point in time to experiment with different renewable layouts and satisfy the demands of the customer. This rapid iteration is used to maintain constant communication with the customer, ensuring that the solution Greengenuity provides is tailored to the needs of the customer. In addition, a model is more easily digestible and visually pleasing than a multipage packet, so it can be used in presentations or to show the general public the city's plan.

XII. Partnerships

In order to supply the necessities needed to run on 100% renewable energy, Greengenuity is able to connect Providence with companies to procure the resources needed to reach this goal. In order to implement infrastructure for renewable power generation, Greengenuity plans to partner with local manufacturers of solar panels, wind turbines, and other related equipment.

A. American Superconductor (WindTec - AMSC)

A wind turbine manufacturer based in Ayer, Massachusetts (American Superconductor), who could partner with Greengenuity to provide wind turbines for the client at a reduced rate.

B. SolSmart

A distinguished company that aids cities in the adoption of solar energy by providing no-cost technical assistance, that we also have an intention in partnering with. This company aids to reduce costs for solar customers, save local governments time and money, and improve business prospects for solar companies. With SolSmart, Providence makes it easier for residents and businesses to go solar by reducing solar "soft-costs" and barriers, including those related to zoning, planning, permitting, inspection, and customer acquisition.

C. First Solar

An American solar panel manufacturer that we partner with. They boast low prices and top-notch technology, and specialize in large-scale solar installations.

D. Ambri

A company that specializes in liquid metal batteries, combining technological innovation with commercial applicability to solve the biggest problems and has raised over $70 million in equity financing since its founding in 2010. Ambri stores the energy gathered from wind and solar efficiently, inexpensively and reliably. Their technology fundamentally changes the way power grids operate, increasing the contribution from renewable resources and reducing the need to build traditional power plants.

E. Verdigris

An AI-centered company based out of California whose main goal is to eliminate electricity waste. This company just so happens to be our employer sponsor, but we do find their technology extremely useful. Although they are not a local company to Providence, we believe the addition of their advanced sensory technology could not only help Providence save electricity, but also expand Providence's experience with new innovative tech in the clean energy industry.

XIII. References

Providence:

1. "Providence, RI Electricity Rates." *Electricity Local*. Web. 05 Aug. 2021.
2. "Energy Report 2018." Web.
3. "Open Data Portal." *Socrata*. Web. 05 Aug. 2021.
4. "List of Power Stations in Rhode Island." *Wikipedia*. Wikimedia Foundation, 08 July 2021. Web. 05 Aug. 2021.
5. "Do you know where your power comes from?" *National Grid* Aug. 2017

Rhode Island Renewable Energy and Funds:

1. "Renewable Energy Fund: REF Grants, Programs & Materials." *Rhode Island Commerce*. 04 Aug. 2021. Web. 05 Aug. 2021.
2. "What Is Community Solar." *What Is Community Solar " Rhode Island Community SOLAR MARKETPLACE*. Web. 05 Aug. 2021.
3. "Commercial Property Assessed Clean Energy." *RIIB*. Web. 05 Aug. 2021.
4. "State of Rhode ISLAND: Office of Energy Resources." *Rhode Island -Office of Energy Resources*. Web. 05 Aug. 2021.
5. "City of Providence Recognized by U.S. Department of Energy FOR Solar-friendly Policies." *City of Providence*. 26 June 2017. Web. 05 Aug. 2021.
6. "State of Rhode ISLAND: Office of Energy Resources." *Rhode Island -Office of Energy Resources*. Web. 05 Aug. 2021.
7. "Brownfields 2017 Area-Wide Planning Grant Fact Sheet Providence, RI." *EPA*. Environmental Protection Agency. Web. 05 Aug. 2021.

Solar:

1. "How Does Solar Work?" *Energy.gov*. Web. 05 Aug. 2021.
2. User, Guest. "R.I. Has Options for 100 Percent Renewable by 2030." *EcoRI News*. EcoRI News, 25 Nov. 2020. Web. 05 Aug. 2021.
3. "What Is Community Solar." *What Is Community Solar " Rhode Island Community SOLAR MARKETPLACE*. Web. 05 Aug. 2021.
4. "Cumberland Solar PV." *EnergySage*. Web. 05 Aug. 2021.
5. "Solar Siting Opportunities for Rhode Island." *Synapse Energy Economics, Inc.* Web.
6. "Rhode Island Office of Energy Resources Clean Energy Grant Incentive Program." *State of Rhode Island Office of Energy Resources*. Web.

7. "Cost of Commercial Solar Panels & Installation." *Freedom Solar*. 22 Apr. 2021. Web. 05 Aug. 2021.
8. "100Kw Solar System: Commercial Solar." *Infinite Energy*. Web. 05 Aug. 2021.

Wind:

1. "Wind Turbine COST: Worth the Million-dollar Price in 2021?" *Weather Guard Lightning Tech*. 22 July 2021. Web. 05 Aug. 2021.
2. The Brattle Group. "The Road to 100% Renewable Electricity by 2030." *Rhode Island TECHNICAL SUPPORT DOCUMENT*. Mar. 2021. Web.
3. "How Gas Turbine Power Plants Work." *Energy.gov*. Web. 05 Aug. 2021.
4. Kuffner, Alex. "Offshore Wind Developers Announce PROVPORT FACILITY." *The Providence Journal*. The Providence Journal, 14 Apr. 2021. Web. 05 Aug. 2021.
5. "State of Rhode ISLAND: Office of Energy Resources." *Home- Rhode Island -Office of Energy Resources*. Web. 05 Aug. 2021.
6. Suburbanstats.org. "Current Providence, Rhode Island POPULATION, Demographics and Stats in 2020, 2019." *SuburbanStats.org*. Web. 05 Aug. 2021.
7. "Resources and FAQs." *Offshore Wind Connecticut and Rhode Island*. Web. 05 Aug. 2021.

Biofuels:

1. Dilthey, Max Roman. "How Does Biofuel Work?" *It Still Runs*. 10 Jan. 2019. Web. 05 Aug. 2021.
2. "Biomass Energy Costs." *Help Save Nature*. 07 Aug. 2010. Web. 05 Aug. 2021.

Energy Storage:

1. Dahal, Sahas, and Filip Poutintsev. "Pros and Cons of Lithium-Ion Batteries." *Honest Pros and Cons*. 12 Mar. 2021. Web. 05 Aug. 2021.
2. Writer, Published by, and Name *. "The Pros and Cons of Lithium-Ion Batteries." *Remis Power Systems Inc.* 08 Apr. 2020. Web. 05 Aug. 2021.
3. Jacoby, Mitch. "It’s Time to Get Serious about Recycling Lithium-Ion Batteries." *C&EN*. Web. 05 Aug. 2021.
4. "Pumped Storage Hydroelectric Power Plants: Issues and Applications." *Energy Regulators Regional Association*. Web.
5. "How Compressed Air Is Used for Renewable Energy." *Quincy Compressor*. 19 Feb. 2021. Web. 05 Aug. 2021.

Partnerships:

1. "City of Providence Recognized by U.S. Department of Energy FOR Solar-Friendly Policies." *City of Providence*. 26 June 2017. Web. 05 Aug. 2021.
2. "This Battery Will Change the World." *Ambri*. Web. 05 Aug. 2021.
3. “Smart Building Management.” Verdigris, verdigris.co/.
4. “First Solar: A Leading Global Provider of COMPREHENSIVE PV Solar Solutions.” Leading Global Provider of Comprehensive PV Solar Solutions | First Solar, www.firstsolar.com/.
5. “Nationally Distinguished. Locally Powered.” *SolSmart*, solsmart.org/.
6. “American Superconductor (AMSC).” *AMSC*, 10 Apr. 2019, www.amsc.com/.

Energy Conservation Methods:

1. Clements, Joe. "19+ Energy Conservation Methods: Eco-Friendly Ways to Reduce Energy." *GreenCoast*. 28 July 2021. Web. 05 Aug. 2021.
2. “Smart Building Management.” Verdigris, verdigris.co/.